

Laboratory guide nº 2 – Performance evaluation

Objectives

The aim of this laboratory work is to study discrete-time Markov chains (DTMCs), continuous-time Markov chains (CTMCs), queuing systems, and their applications.

Exercises

A. Discrete-time Markov chains (class 5)

Before performing these exercises study the videos “Stochastic processes” (stocproc) and “Discrete-time Markov chains” (dtmc).

1. The script `dtmc_bernoulli_plot.R` compares sample paths of the Bernoulli and 2-DTMC processes, while ensuring that the success probability of the Bernoulli process equals the average of the 2-DTMC process, which is $(1 - \alpha)/(1 - \alpha + \beta)$. Try different values of α and β (at least two sets of values) leading to significantly different behavior. What type of behavior can 2-DTMC processes model that Bernoulli processes can't? **Note:** Include in your report the jpeg files saved by the `dtmc_bernoulli_plot.R` script.
2. Suppose you are given a dataset corresponding to the sequence of daily attacks to a server, where “0” means “not attacked” and “1” means “attacked”; this dataset is in file “2dtmcddata.txt”. Suppose also that this type of attack can be well modelled by a 2-DTMC. Write an R script that imports the dataset and estimates the transition probabilities of the 2-DTMC. To import the file, use the R command `mydata=read.table("2dtmcddata.txt",sep="\t")`, which imports the dataset into a data frame called `mydata`. What are the estimates of the various transition probabilities? **Note:** Present the results in a table.
3. Consider the following model to compute the throughput of slotted ALOHA with a finite number of users: there are N users; time is slotted and the duration of one slot is the time to transmit a packet; each user can be in one of two states: THINKING or BACKLOGGED; THINKING users transmit one packet in a slot with probability σ and change to the BACKLOGGED state if they collide; BACKLOGGED users have exactly one packet waiting and transmit in a slot with probability p ; if two or more users transmit in the same slot there will be a collision; collided users are placed in the BACKLOGGED state (if not already there); if only one user transmits in a slot, the transmission will have success and the user returns to the THINKING state (if not already there); note that a user that transmits in the THINKING state remains in this state if the transmission was successful, and a user that transmits in the BACKLOGGED state remains in this state if the transmission collided.

- a. The script `aloha_plot.R` uses the function `aloha_theo()` to plot the theoretical throughput as a function of σ , for different values of N and p . The script is prepared to plot two graphs: one for $N = 10$ and $p = 0.3, p = 0.4, p = 0.5, p = 0.6$, and another for $N = 25$ and $p = 0.2, p = 0.3$. How do you explain the evolution of the throughput with the parameters N, p , and σ ? **Note:** Include in your report the jpeg files saved by the `aloha_plot.R` script.
- b. Build a script to estimate the throughput using discrete event simulation. How do these results compare with the theoretical ones? The comparison must be done for at least 3 sets of (N, σ, p) values that lead to significantly different throughput values. **Notes:** Present the results in a table. Do not be literal in your code: there is no need to model the state of each user! In the report, explain briefly the simulation code. The theoretical throughput can be obtained through the `aloha_theo()` function.
- c. How does ALOHA compare with CSMA and CSMA/CD in terms of throughput? Identify technologies that use ALOHA, CSMA, and CSMA/CD, e.g., by searching on the Internet.

B. Discrete-time Markov chains and Continuous-time Markov chains (class 6)

Before performing these exercises, study the videos “Discrete-time Markov chains” (dtmc), “Exponential distribution” (expdist), “Poisson process” (poiproc), and “Continuous-time Markov chains” (cmtc).

4. Consider the DTMC with 3 states represented in the next figure. Obtain the limiting state probabilities using three different approaches: (i) solving the balance equations, (ii) using matrix multiplication, and (iii) simulation. How do the results compare? **Notes:** Present the results using a table. Use the `expm` package to compute matrix powers; in this case, the n -th power of matrix A is computed as `A %^% n`. In the report, explain how you did the calculations and simulations.

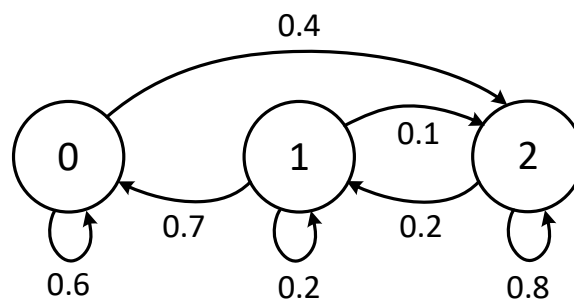


Figure 1: DTMC.

5. Consider the CTMC with 3 states represented in the next figure, where $\lambda_{21} = \lambda_{23} = \lambda_{31} = \lambda_{32} = 1$ and $\lambda_{12} = 2$. Simulate the CTMC to estimate the limiting state probabilities and compare with the theoretical results. The CTMC simulation must be performed in two different ways, one according to view 1 and another according to view 2 of the CTMC definition. **Note:** Use the `solve()`

function to determine the theoretical limiting probabilities. In the report, explain how you did the calculations and simulations.

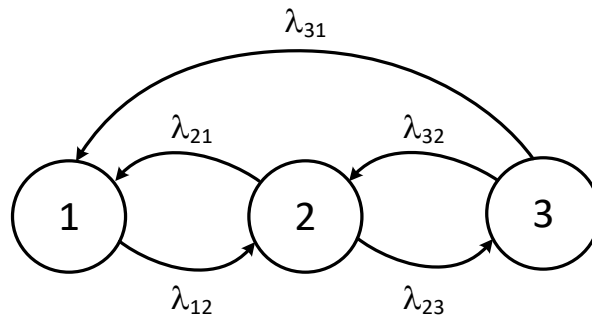


Figure 2: CTMC.

C. Queuing systems (class 7)

Before performing these exercises study the video “Queuing systems” (queuesys).

6. The aim of this exercise is to study the consequences of the lack of stability in queuing systems. Use the M/M/1 simulator that estimates the average queuing delay. Consider two values of the system load: $\rho = 1$ and $\rho = 2$. For these two values, study the average queuing delay as a function of the simulation run length (consider at least three values of NumQueueCompleted). What can you conclude? Is the system stable?
7. Using the ppl.R simulator:
 - a. Simulate a point-to-point link with FIFO scheduling, which serves two flows, both with Poisson arrivals and exponentially distributed packet sizes. The simulations must be executed for two values of λ/μ that lead to significantly different performance values (consider only stable conditions). Compare the values obtained with the theoretical ones and draw conclusions.
 - b. Simulate a point-to-point link with FIFO scheduling, which serves two flows, both with Poisson arrivals and fixed packet sizes. The simulations must be executed for two values of λ/μ that lead to significantly different performance values (consider only stable conditions). Compare the values obtained with the theoretical ones and draw conclusions. How does the performance compare between exponentially distributed packet sizes (question a) and fixed packet sizes? What can you conclude regarding the impact of the service time distribution on the performance of these queuing systems?
 - c. Simulate a point-to-point link with strict priority scheduling, which serves two flows with different priorities, both with Poisson arrivals and fixed packet sizes. The simulation must be executed for a set of parameters $(\lambda_1, \lambda_2, \mu)$ that leads to a significant difference in performance values between the two flows; specifically, the average delay of the higher priority flow must be at least 4 times smaller than the average delay of lower priority flow; consider only stable

conditions. Compare the values obtained with the theoretical ones and draw conclusions.

In each case above, indicate the parameters used in the simulations, including the simulation time. **Note:** *The ppl simulator includes several scripts; the parameters are edited in the parameters.R script, and this is the only script that you need to modify.*

D. Workloads (class 8)

Before performing this exercise, study the video “Workloads” (workloads).

8. The aim of this exercise is to study the impact of the workload variability on the performance of queuing systems. Consider two M/G/1 systems with bimodal (mice-elephant) service distribution. The first system is characterized by the set of parameters ($S_1 = 1, p_1 = 0.9, S_2 = 11, p_2 = 0.1$) and the second by ($S_1 = 1, p_1 = 0.99, S_2 = 101, p_2 = 0.01$), where S_i is the service time of job i and p_i its probability. The script mg1_bimodal.R addresses this problem, using both discrete event simulation and queuing theory.
 - a. Compare the average queuing delay of the M/M/1 and of the two M/G/1 systems described above. What is the impact of the workload variability in this performance metric? **Note:** *Present the results in a table.*
 - b. Compare the sample variance of the average queuing delay estimates obtained in the three cases of previous exercise (use 100 samples to compute the variance). What can you conclude regarding the impact of the job size variability in simulation results? **Note:** *Present the results in a table.*
 - c. Find a way of improving the average queuing delay of the M/G/1 systems. **Note:** *Mice and elephants should be treated differently.*

E. Server farms (class 8)

Before performing this exercise, study the video “Server farms” (servfarm).

9. Consider a Web server farm with two servers. The script webserverfarm.R estimates the average delay in system under two different task assignment policies: random and Join Shortest Queue (JSQ). The script assumes that $\mu=1$, so always use this value. Compare the two policies for different values of the system ρ (at least two values). What can you conclude?
10. Investigate the load balancing algorithms used by NGINX, described [in this page](#). What is NGINX? Which algorithms have better performance in terms of average response delay?

F. Packet switched networks (class 9)

Before performing these exercises, study the video “Packet switched networks” (pktnet).

11. Write an R script to calculate (i) the average packet delay of each flow, (ii) the average number of packets in the network, and (iii) the average packet delay in the network, of a packet switched network using the Kleinrock approximation. Your script must be generic, i.e., it must accept any network topology and any set of flows. Test your script using simple examples, for which hand calculations are easily performed, and report the results of your tests. The network links should be identified using contiguous integers, starting at 1. **Note:** Use the same input data structures as the *pnet.R* simulator, namely the *LinkCapacities* vector and the *Flows* list of lists.
12. Consider a simple packet switched network comprised of two tandem links with capacity 64 Kb/s. A flow where packets arrive according to a Poisson process and the size of packets is exponentially distributed with mean 1000 bits is offered between the first node and the last node of the network. Compare the average packet delay obtained (i) through the Kleinrock approximation and (ii) through simulation (using the *pnet.R* simulator), for four distinct values of $\rho = \lambda/\mu$ namely 0.05, 0.5, 0.95 e 0.975. What can you conclude regarding the accuracy of the Kleinrock approximation? **Note:** Present the results in a table.
13. Consider the packet switched network of Figure 3. All links have 256 Kb/s of capacity. The network is initially serving four packet flows using non-bifurcated fixed routing: flow 1 with route 1-2-4-6 and $\lambda_1 = 215$ packets/s; flow 2 with route 1-3-5 and $\lambda_2 = 64$ packets/s; flow 3 with route 1-3-5-6 and $\lambda_3 = 128$ packets/s; flow 4 with route 2-5 and $\lambda_4 = 128$ packets/s. Assume that the flows are characterized by Poisson arrivals and exponentially distributed packet sizes with mean 1000 bits.
 - a. Using the Kleinrock approximation, calculate the average network packet delay.
 - b. Confirm the results obtained in a) using simulation (utilize the *pnet.R* simulator). Indicate the input parameters of the simulations, including the simulation time (contents of *parameters.R* script).
 - c. Consider now that flow 1 (and only flow 1) can be bifurcated through routes 1-2-4-6 and 1-2-5-6. Determine the optimal bifurcation of flow 1 (using hand calculations) and compare the average network packet delay with the one of question a).
 - d. What if all flows can be bifurcated? Can the average network packet delay still be improved? Use the *bg.R* script to answer this question (it implements the gradient projection algorithm). Indicate input the parameters of the algorithm (contents of *parameters.R* script).

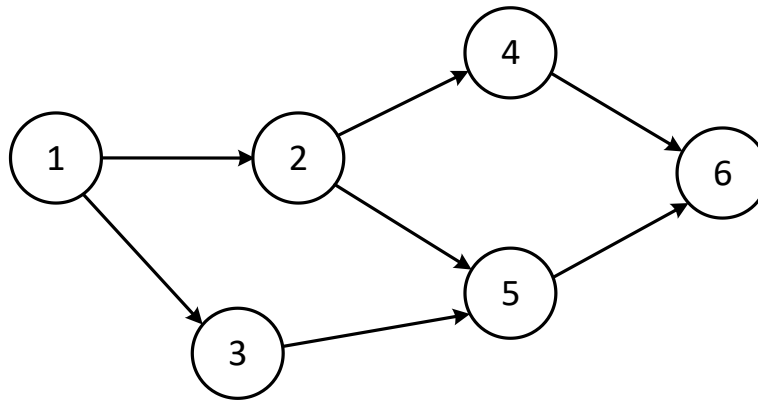


Figure 3: Packet switched network.

G. Circuit switched networks (class 10)

Before performing this exercise, study the video “Circuit switched networks” (circuitnet).

14. Consider the circuit switched network of Figure 4. The network has 4 nodes, 5 links and 3 call flows:

- Flow 1 from node 1 to node 2 with offered load $\rho = 0.5$.
- Flow 2 from node 2 to node 3 with offered load $\rho = 0.5$.
- Flow 3 from node 1 to node 3 with offered load $\rho = 0.5$.

Each call requires a capacity of $b = 64$ Kb/s. You are only allowed to install capacity at the links represented in the figure, in multiples of 64 Kb/s. Each link has a different operational cost per 64 Kbit/s circuit:

- Links 1-2 and 2-3: 500 Euros per 64 Kbit/s circuit
 - Link 2-4: 1000 Euros per 64 Kbit/s circuit.
 - Links 1-4 and 4-3: 100 Euros per 64 Kbit/s circuit.
- a. Determine (i) the routes of the 3 flows and (ii) the capacities of the 5 links that minimize the total cost of the installed solution, such that the blocking probability of each flow is less than 1%. Note that some links may have no capacity installed. Obtain this solution using the `reducload_cost.R` script, which computes the flow blocking probabilities using the reduced load approximation. **Note:** Use a trial-and-error approach guided by the link probabilities which are output by the `reducload_cost.R` script.
 - b. Once the optimal solution is found, use the `cnet.R` simulator to confirm the results. Which are more trustful, the results obtained through the reduced load approximation, or the ones obtained through simulation? Indicate the input parameters of the simulations, including the simulation time (contents of `parameters.R` script).
 - c. Obtain the flow blocking probabilities using the product bound (script `prodbound.R`). How do the results compare with the previous ones?

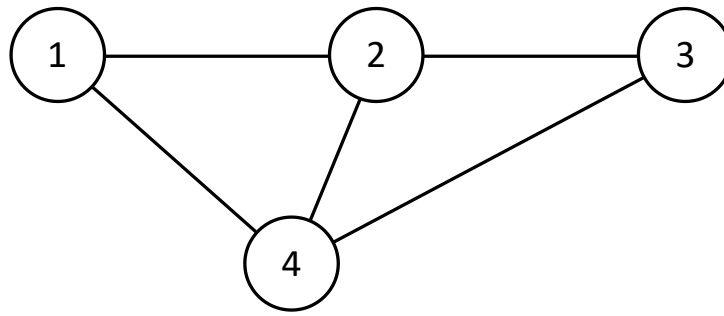


Figure 4: Circuit switched network.

References

- DDRS course notes, chapters 2, 4, 5, and 6, Rui Valadas (see Fenix)
- Sheldon M. Ross, "Introduction to Probability Models", 10th edition, Academic Press, 2010.
- Mor Harchol-Balter, "Performance Modeling and Design of Computer Systems", Cambridge University Press, 2013.
- R. Rom, M. Sidi, "Multiple Access Protocols: Performance Analysis", Springer-Verlag, 1990.
- D. Bertsekas, R. Gallager, "Data Networks", 2nd edition, Prentice-Hall, 1992.
- K. Ross, "Multiservice Loss Models for Broadband Telecommunication Networks", Springer-Verlag, 1995.
- S. Keshav, "An Engineering Approach to Computer Networking: ATM Networks, the Internet, and the Telephone Network", Addison-Wesley, 1997.

Duration

See the DDRS roadmap.

Work guidelines

Since you already learned how to analyze simulation output data using confidence intervals, all simulation results must be presented with confidence intervals.

The code that you develop in R must be well structured, simple, and easy to read. You must insert comments in your code to improve readability.

Visualization

All exercises must be demonstrated to the teacher during the classes.

Report

See the DDRS roadmap for instructions on how to write the report.

Plagiarism

Any copy detected between reports leads to cancelling all the involved reports, and a very angry teacher.

Extra (optional) exercises

15. The aim of this exercise is to study the PASTA property. You will address two systems: M/M/1 and G/M/1 where the input process is deterministic (constant interarrival times). The aim is to compare the proportions of times that an arrival finds 0, 1, or 2 clients in the system, with the limiting probabilities of having 0, 1, or 2 clients in the system. Start your work by completing the script `pasta_mm1_inc.R`, which addresses the M/M/1 system. Then, modify the previous script to address the G/M/1 system with constant interarrival times. Use the same values of λ and μ in both systems; we suggest $\lambda = 1$ and $\mu = 2$. What can you conclude regarding the PASTA property?
16. Heavy-tailed distributions have been observed many times in the context of computer networks and systems. In this exercise, you will study the M/G/1 queuing system with Bounded Pareto service distribution and compare it with the M/M/1 queuing system. This is addressed in the script `mg1_boundedpareto.R`. You will consider two sets of parameters for the Bounded Pareto distribution: $(L = 0.297, H = 10^3, \alpha = 1.4)$ and $(L = 0.286, H = 10^6, \alpha = 1.4)$.
- Compare the average queuing delay of M/M/1 and of the two M/G/1 systems described above. What is the impact of the workload variability in this performance metric?
 - Compare the sample variance of the average queuing delay estimates obtained in the three cases of previous exercise (use 100 samples to compute the variance). What can you conclude?
17. The aim of this exercise is to study the impact of the traffic burstiness in the performance of queuing systems. Suppose that the traffic offered to a single server queue with finite capacity and exponentially distributed service times is an Interrupted Poisson Process (IPP), i.e., it has a bursty nature. This is an IPP/M/1/K queuing system. There is no closed form solution for the loss probability of this system. Thus, we may be tempted to approximate this system by another one that has a closed form solution, e.g., the M/M/1/K queuing system. This problem is addressed through simulation in the script `ippm1k.R`. The script estimates the loss probability of the two systems, assuming that the average arrival rate is the same in both. The script input parameters are the peak arrival rate λ_1 (i.e., the arrival rate in state 1), the probability of state 1 π_1 , the system utilization $\rho = \lambda/\mu$, the queue size, i.e., $K - 1$, and the ratio between the peak rate and the output rate of state 1 $R = \lambda_1/q_{10}$. From these parameters, the script calculates the average arrival rate, $\lambda = \pi_1 \lambda_1$, the service rate $\mu = \lambda/\rho$, the output rate of state 1 $q_{10} = \lambda_1/R$ and the output rate of state 0 $q_{01} = \pi_1 q_{10}/(1 - \pi_1)$. Try to answer the following questions. How does the loss probabilities evolve as a function of K , R , and ρ in the IPP/M/1/K system? What is the relative error of assuming an M/M/1/K model for an IPP/M/1/K system?
18. Write an R script to simulate two queues served by the DRR (Deficit Round Robin) scheduling algorithm. The packet arrivals must be according to a

Poisson process in each queue, with different rates for each queue. The packet sizes must be uniformly distributed in each queue, with different averages for each queue. The quantum values of each queue must be selected considering the computational effort in deciding which queue must transmit a packet. The output data must be the throughput of each queue (in bits/s). Your implementation can be based on the script `rand_sched.R`, which simulates a random scheduling discipline. You must convincingly demonstrate that your implementation is correct by studying the throughput of each queue for different (and well thought out) combinations of the input parameters. In the report, include a brief description of the changes made to the `rand_sched.R` script.

Notes:

- *Try to understand the `rand_sched.R` script before doing the exercise.*
- *Include a vector with the quantum values as an input parameter.*
- *Regarding events, only the departure event needs to be modified.*
- *Transmitting a packet may require circulating several times between the queues, until enough credit is gained.*
- *Make sure to reset to zero the credit of a queue found empty.*
- *If there is at least one packet queued, that packet will have to be transmitted (DRR is work conserving). For that, you may need to circulate several times between the queues, until enough credit is granted. This can be implemented through a `while (TRUE)` loop.*
- *To test your implementation of the DRR scheduler, the link must be overloaded, i.e., the total offered load must be significantly higher than the link capacity. If the total offered load is lower than the link capacity, each flow will get a throughput equal to its offered rate, and the scheduler will not be able to differentiate among flows. Remember that DRR is work-conserving. Note that when comparing the link capacity with the offered load the units must be the same. If the link capacity is in bits/s, the offered load must also be in bits/s. For the offered load this implies, for each flow, multiplying the arrival rate (in packets/s) by the average packet size (in bits/packet).*

19. Analyze the information contained [in this page](#), relative to the QoS mechanisms of 7x50 Nokia routers. What is the packet scheduling discipline used for dequeuing? Explain briefly the four QoS components.